



Industrial Motor

DC Motor

Model: FM-5024C

OPERATION AND MAINTENANCE MANUAL

Version: 1.9

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For technical support or additional information, please contact your local distributor or visit our website.

Before operating the FM-5024C, ensure that all installation requirements have been met and that the equipment has been properly commissioned by qualified personnel.

This manual provides comprehensive information for the operation, maintenance, and troubleshooting of the Industrial Motor model FM-5024C.

Please read this manual thoroughly before installation and operation. Keep this manual in a safe place for future reference.

This document contains detailed technical specifications, installation procedures, operational guidelines, and maintenance schedules to ensure optimal performance and safety.

This equipment must be installed and operated in accordance with local codes, regulations, and industry standards. Consult with qualified personnel if you have questions about compliance requirements.



Note

Keep this manual accessible at the equipment location for quick reference during operation and maintenance activities.

For more information see:

- **Maintenance Schedule - Inspection Requirements**

Warning	Description	Elimination/Action
	Allow equipment to cool before maintenance. Use appropriate protective equipment when handling hot components.	Wait for equipment to reach ambient temperature. Wear heat-resistant gloves.
	Some lubricants and materials may cause allergic reactions in sensitive individuals.	Use protective gloves. Ensure adequate ventilation. Consult material safety data sheets.
	Pressure can build up during operation. Sudden release can cause injury.	Release pressure slowly using appropriate procedures. Never remove caps or plugs under pressure.
	Overfilling can cause leakage, overheating, and equipment damage.	Follow manufacturer specifications for fill levels. Check levels regularly.
	Mixing incompatible oils can cause equipment failure and void warranty.	Use only specified lubricants. Drain completely before changing oil type.
	Warming oil before use improves flow and ensures proper distribution.	Heat oil to recommended temperature (typically 40-50°C) before application.
	The amount of new lubricant should match the volume drained to maintain proper levels.	Measure drained volume accurately. Add equivalent amount of new lubricant.
	Residual oil can be slippery and cause falls. Dispose of properly.	Clean up spills immediately. Use appropriate absorbent materials. Follow local disposal regulations.

Specifications may vary slightly due to manufacturing tolerances and normal variations in materials. Contact technical support if you require exact specifications for your application.

All specifications are subject to change without notice as improvements are made to the product. Always refer to the latest version of this manual for current specifications.

Some specifications may be optional or configurable. Consult with technical support or your local distributor to determine available options for your specific requirements.

General Specifications

Parameter	Value	Unit	Notes
Power Range	15	kW	Optional
Voltage Range	690V	V	Optional
Frequency	50	Hz	Optional
Speed Range	1500	rpm	Optional
Efficiency Class	IE3		As specified
Weight	3538	kg	Approximate
Operating Temperature	-14 to 60	°C	Continuous
Storage Temperature	-28 to 65	°C	
Humidity	12 to 95	% RH	Non-condensing
Protection Class	IP65		IEC 60529
Noise Level	62	dB(A)	At 1m distance
Vibration	≤ 4	mm/s	RMS
Service Factor	1.15		

Dimensions

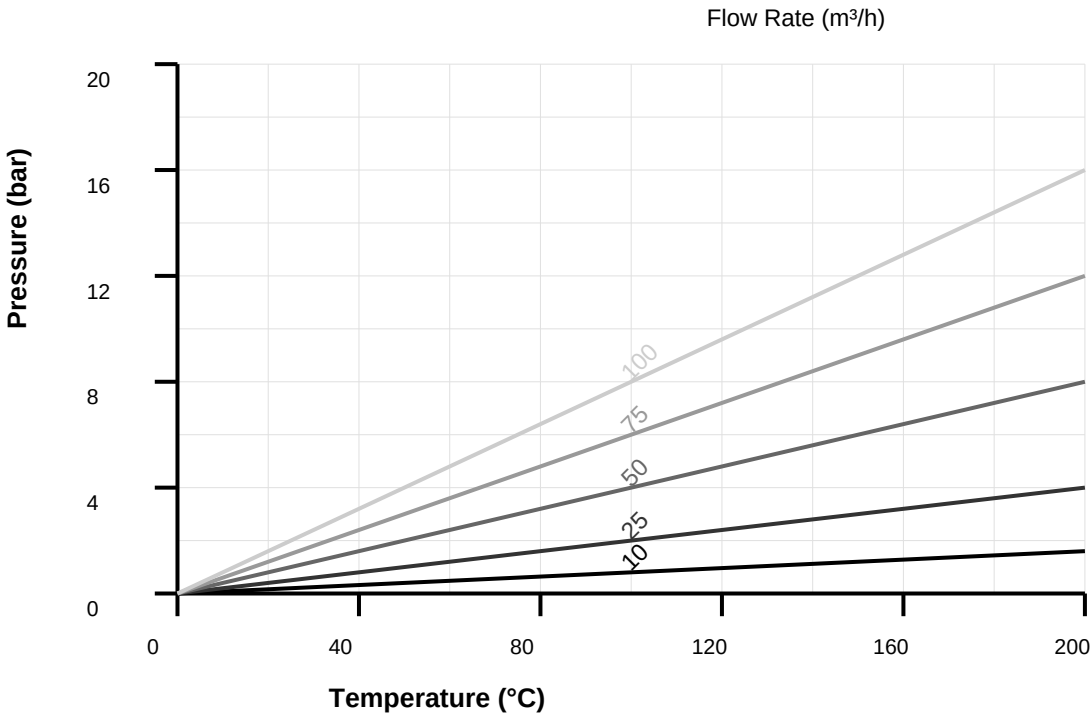
Dimension	Value	Unit	Tolerance
Length	636	mm	±2 mm
Width	791	mm	±2 mm
Height	583	mm	±2 mm
Mounting Holes	M12		
Shaft Diameter	62	mm	±0.1 mm
Base Height	58	mm	±1 mm

Performance Data

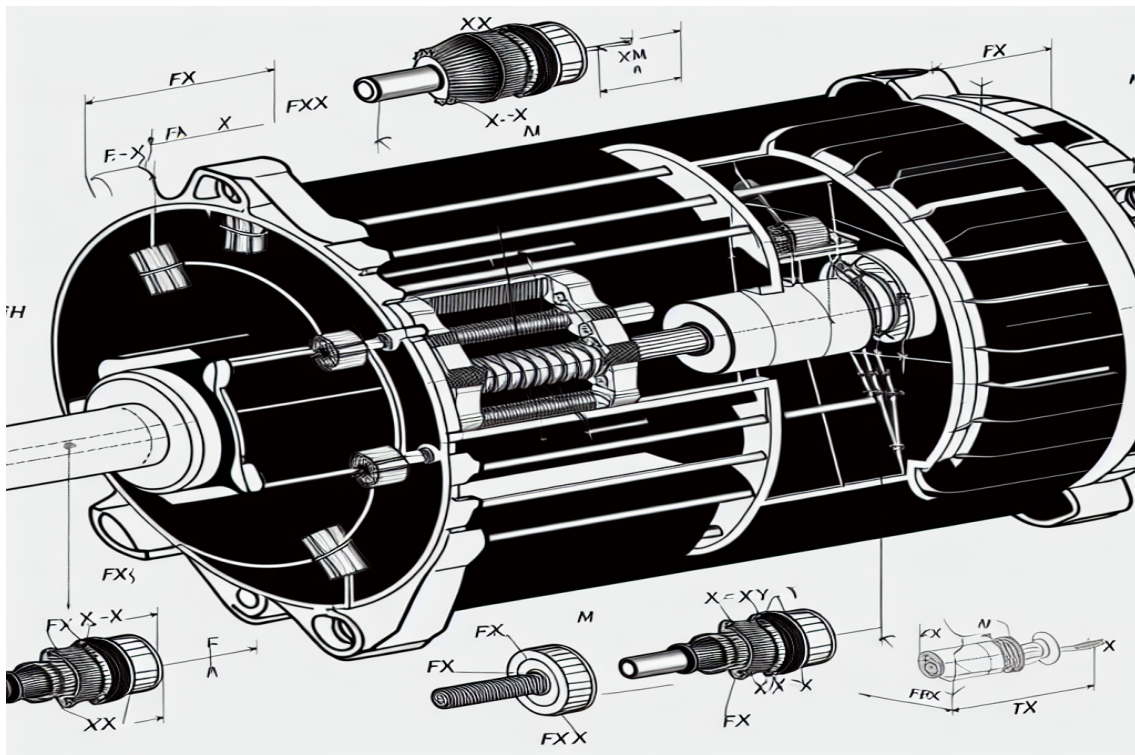
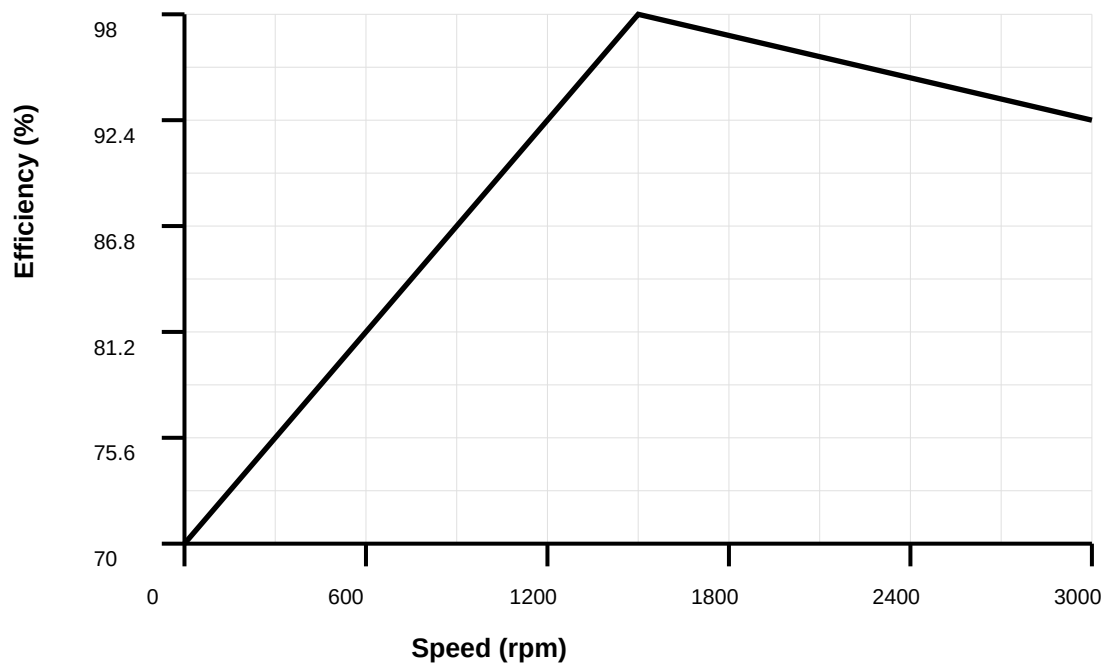
Operating Condition	Minimum	Nominal	Maximum	Unit
Power Consumption	83	97	105	%
Efficiency	87	92	97	%
Load Factor	22	78	102	%
Speed	742	1475	2958	rpm
Torque	26	80	176	Nm

Environmental Conditions

Condition	Operating Range	Storage Range	Unit
Temperature	-19 to 56	-23 to 69	°C
Relative Humidity	14 to 85	6 to 91	%
Altitude	0 to 1680	0 to 2050	m
Vibration	≤ 2	≤ 6	mm/s



Efficiency vs Speed



Electrical Specifications

Parameter	Value	Unit	Standard
Supply Voltage	230V	V AC	IEC 60038
Frequency	50 Hz	Hz	IEC 60034
Current Rating	96	A	
Power Factor	0.90		
Starting Current	567	% of rated	
Insulation Class	H		IEC 60085
Protection	IP66		IEC 60529

Electrical connections must be made in accordance with local electrical codes and manufacturer specifications. Use only qualified electricians for electrical work.

Protection class ratings indicate the degree of protection against ingress of solid objects and water. Ensure that the protection class is appropriate for your installation environment.

4.1 Installation Requirements

Installation Procedures

Action	Note
• Check the sealing to ensure that: • The sealing is of the correct type. • There is no damage on the main lip.	Article number is specified in Equipment on page 166.
• Lubricate the sealing with grease just before fitting. • (Not too early - there is a risk of dirt and foreign particles adhering to the sealing.) • Fill 3/4 of the space between the dust lip and the main lip with grease. • If the sealing is without dust lip, just lubricate the main lip with a thin layer of grease.	Ensure that no grease is applied to the marked surface. Use only approved lubricants specified in the parts list.
• Apply proper torque to all fasteners. • Use torque values specified in the technical data sheet (29 Nm for main bolts). • Tighten in the correct sequence as shown in the installation diagram.	Overtightening can damage components. Undertightening can cause loosening during operation.
• Inspect the shaft surface before mounting. • If scratches or damage are found, the shaft must be replaced since it may result in future leakage. • Do not try to grind or polish the shaft surface to get rid of the defect.	Ensure proper surface finish is maintained. Refer to technical specifications for surface roughness requirements.



CAUTION

Please observe the following before commencing any assembly or maintenance of the motor:

- Protect the motor windings during transport and mounting. Avoid exposure to moisture and contaminants.
- Ensure the motor is completely de-energized and locked out before performing any maintenance work.
- The installation of bearings and seals must be carried out in a clean environment free from dust and debris.
- Use appropriate lifting equipment rated for the motor weight. Never lift by the shaft or terminal box.
- Do not apply excessive force when mounting couplings. Use proper alignment tools and follow manufacturer specifications.

Environmental factors such as temperature, humidity, and exposure to contaminants must be considered during installation planning.

After installation, perform all required alignment, calibration, and verification procedures before placing the equipment into service.

Electrical connections must be made by qualified electricians in accordance with local electrical codes and manufacturer specifications.

Proper installation of the Industrial Motor is critical for safe and reliable operation. This section provides detailed procedures and requirements for installation.

Before installation, ensure the following requirements are met:

Installation Checklist

Requirement	Specification	Status
Adequate space for installation and maintenance access	As per installation manual	■
Proper foundation or mounting surface	As per installation manual	■
Correct electrical supply voltage and frequency	As per installation manual	■
Appropriate environmental conditions (temperature, humidity)	As per installation manual	■
Compliance with local building and electrical codes	As per installation manual	■

Foundation Requirements

Parameter	Minimum	Recommended	Unit
Foundation Thickness	306	334	mm
Concrete Strength	C22	C29	
Reinforcement	As per local code	As per local code	
Anchor Bolt Size	M16	M24	
Anchor Bolt Depth	205	223	mm

4.2 Installation Steps

Installation Procedure

Step	Description	Duration	Tools Required
1	Unpack the equipment and inspect for damage.	1 hour	Multimeter, Wrench set, Level
2	Position the equipment according to layout drawings.	1 hour	Multimeter
3	Secure the equipment to the foundation using appropriate fasteners.	2 hours	Torque wrench
4	Connect electrical wiring according to wiring diagrams.	2 hours	Alignment tools, Torque wrench, Multimeter
5	Connect auxiliary systems (cooling, lubrication, etc.).	30 min	Multimeter
6	Perform initial alignment and calibration.	30 min	Alignment tools
7	Verify all connections and safety systems.	30 min	Alignment tools
8	Perform initial test run according to startup procedure.	2 hours	Multimeter, Torque wrench

Torque Specifications

Component	Bolt Size	Torque	Unit
Mounting Bolts	M12	69	Nm
Shaft Coupling	M8	31	Nm
Terminal Connections	M6	10	Nm
Cover Bolts	M8	15	Nm

Document all installation measurements, adjustments, and test results. This information is valuable for troubleshooting and future maintenance.

During installation, periodically verify that all connections are secure and that no damage has occurred to components. Address any issues before proceeding.

Ensure that installation personnel are properly trained and familiar with the equipment. Inexperienced personnel should work under supervision.



Note
Keep installation records including foundation specifications, alignment measurements, and electrical connection details for future reference.
For more information see:
● **Technical Specifications - Torque Values**

5.1 Operating Procedures

The Industrial Motor is designed for continuous operation under normal conditions. Extended operation at maximum capacity may require additional monitoring and maintenance.

Documentation of operating conditions, parameter settings, and performance data is important for optimizing operation and planning maintenance.

Follow the startup sequence exactly as specified. Improper startup procedures may cause equipment damage or safety hazards.

If operating conditions change significantly, review operating parameters and adjust as necessary. Sudden changes may indicate problems requiring attention.

Before operating the equipment, ensure that all installation and commissioning procedures have been completed and that the equipment is in proper working condition.

Regular inspection during operation helps identify potential problems before they become serious. Include visual inspection, parameter monitoring, and listening for unusual sounds.

Follow these procedures for safe and efficient operation:

Operating Parameters

Parameter	Setting	Range	Unit
Startup Sequence	Automatic	Manual/Automatic	
Operating Mode	Continuous	Continuous/Intermittent	
Control Method	Remote	Manual/Automatic/Remote	
Monitoring	Enabled	Enabled/Disabled	
Speed Setpoint	1482	2705	rpm
Load Limit	86	1	%

Startup Sequence

Step	Action	Duration	Status Indicator
1	Power On	Immediate	Power LED
2	System Initialization	14	Init LED
3	Self-Test	27	Test LED
4	Ready State	Continuous	Ready LED
5	Start Command	On Demand	Run LED

5.2 Control Functions

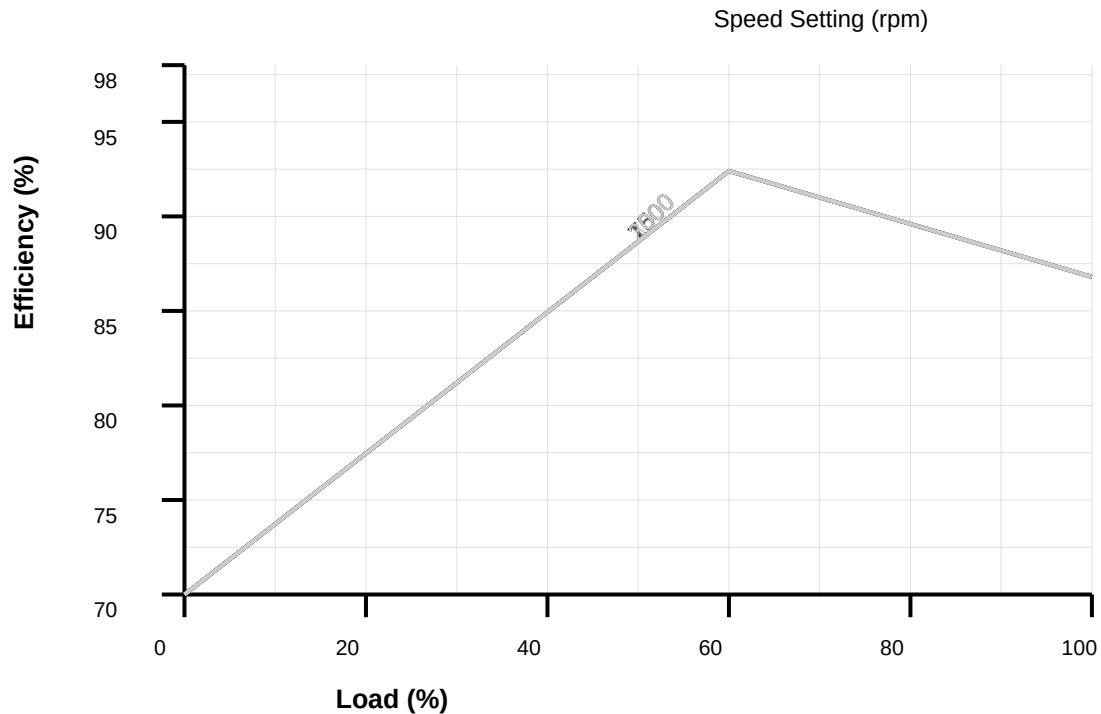
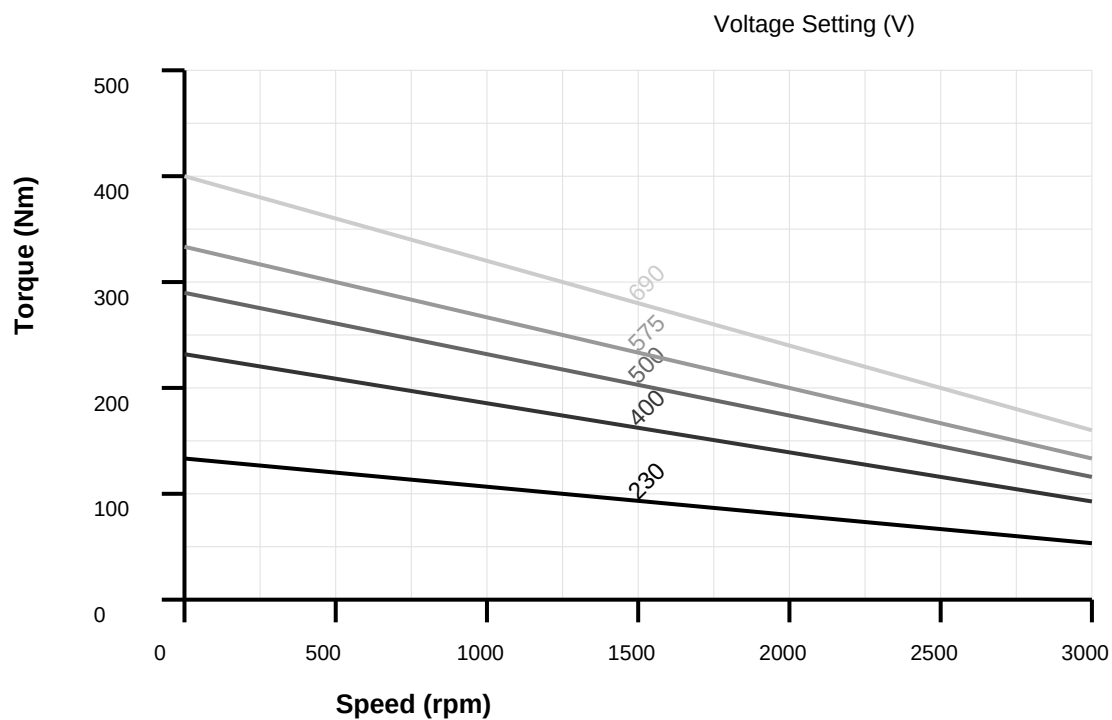
Control Functions

Control	Function	Location	Type
Start/Stop	Controls equipment operation	Panel A	Push Button
Speed Control	Adjusts operating speed within specified range	Panel B	Selector Switch
Emergency Stop	Immediately stops all operations	Panel C	Emergency Stop
Reset	Clears fault conditions and resets system	Panel D	Reset Button
Mode Selection	Selects operating mode	Panel E	Selector

Status Indicators

Indicator	Color	Meaning	Action
Power	Green	System powered	Normal operation
Run	Green	Equipment running	Normal operation
Fault	Red	Fault detected	Check fault code
Warning	Yellow	Warning condition	Review warning
Maintenance	Yellow	Maintenance due	Schedule service

Operating Characteristics



Monitor operating parameters continuously during operation. Significant deviations from normal values may indicate problems requiring attention.

Emergency stop functions must be tested regularly to ensure they are functioning properly. Never disable safety systems.

6.1 Maintenance Schedule

Follow the maintenance schedule strictly to prevent premature failures and ensure optimal performance. Maintenance intervals may need to be adjusted based on operating conditions.

Keep detailed maintenance records including dates, tasks performed, parts replaced, and lubricants used. This information is required for warranty claims and service history.

Use only approved replacement parts and lubricants specified in this manual. Using incorrect parts or lubricants may cause equipment failure and void warranty.

Maintenance Schedule

Maintenance Task	Frequency	Duration	Personnel
Visual Inspection	Daily	5 min	Operator
Lubrication	Every 436 hours	30 min	Technician
Filter Replacement	Every 693 hours	1 hour	Technician
Bearing Inspection	Every 1966 hours	2 hours	Engineer
Complete Overhaul	Every 6729 hours	8 hours	Service Team
Calibration Check	Every 3938 hours	1 hour	Engineer

Inspection Checklist

Item	Check	Acceptance Criteria	Frequency
Bearings	Visual/Temperature	No noise, temp < 70°C	Weekly
Seals	Visual	No leakage	Weekly
Mounting	Visual/Torque	Bolts tight, no movement	Monthly
Electrical	Visual/Measurement	No damage, correct voltage	Monthly
Cooling	Visual/Temperature	Clean, adequate flow	Weekly
Vibration	Measurement	< 5 mm/s RMS	Monthly

6.2 Maintenance Procedures

- Always shut down and lock out equipment before maintenance.
- Follow manufacturer's recommendations for lubricants and spare parts.
- Inspect all components for wear, damage, or contamination.

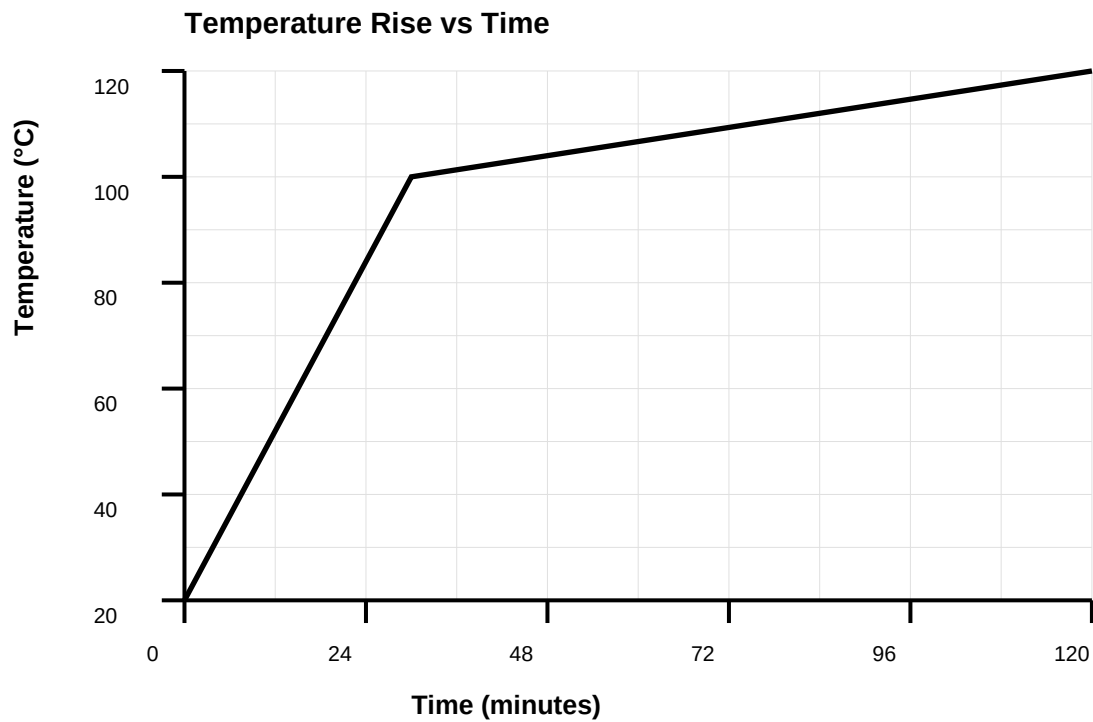
- Replace worn or damaged parts immediately.
- Keep maintenance records for warranty and service history.
- Use only approved replacement parts and materials.

Component Maintenance Intervals

Component	Inspection Interval	Replacement Interval	Condition
Bearings	Every 1610 hours	Every 8167 hours	Based on condition
Seals	Every 680 hours	Every 3492 hours	If leaking
Filters	Every 775 hours	Every 1082 hours	If dirty
Belts	Every 1610 hours	Every 3784 hours	If worn

After completing maintenance, verify that all components are properly installed and that the equipment is ready for operation before restarting.

Temperature Characteristics



Before beginning troubleshooting procedures, ensure the equipment is safely shut down and all safety protocols are followed. Always refer to the safety instructions in Section 2 before performing any diagnostic work.

Troubleshooting the Industrial Motor requires systematic diagnosis of symptoms and their underlying causes. This section provides guidance for identifying and resolving common operational issues.

Many issues can be resolved through basic inspection and maintenance. However, some problems may indicate more serious conditions that require professional service. When in doubt, contact technical support.

Common Issues and Solutions

Symptom	Possible Cause	Solution
Ground fault detected	Insulation failure, moisture ingress, or damaged wiring	Measure insulation resistance, inspect wiring for damage, check for moisture
Encoder feedback error	Encoder failure, cable issue, or mounting problem	Check encoder connections, verify signal integrity, inspect mounting
Excessive current draw	Mechanical binding, short circuit, or motor issue	Check for mechanical obstructions, inspect electrical circuits, test motor
Bearing temperature high	Insufficient lubrication, bearing wear, or misalignment	Check lubrication levels, inspect bearings, verify alignment
Unusual noise during operation	Bearing wear, misalignment, or loose components	Inspect bearings for wear, check alignment, tighten all fasteners
Equipment will not start	Power supply issue or circuit breaker tripped	Check electrical connections, verify voltage, reset circuit breaker if necessary
High power consumption	Mechanical binding, overload, or inefficient operation	Check for mechanical obstructions, verify load, optimize operating parameters
Unusual odors or smoke	Overheating, electrical fault, or lubrication issue	Immediately shut down, inspect for overheating, check electrical connections
Leakage of fluid or lubricant	Damaged seals, gaskets, or overfilling	Inspect seals and gaskets, verify fill levels, replace damaged components
Alarm A14: Vibration level exceeded	Rotor imbalance, misalignment, or bearing failure	Perform dynamic balancing, check alignment, inspect bearings
Temperature rise above normal	Insufficient cooling, high ambient, or internal fault	Check cooling system, verify ambient conditions, inspect internal components

When troubleshooting, always start with the simplest possible causes before investigating more complex issues. Many problems can be resolved through basic inspection and cleaning.

Use systematic diagnostic procedures: observe symptoms, identify possible causes, test hypotheses, and verify solutions. Document each step for future reference.

Part Number	Description	Quantity	Material	Supplier Part No.
FM-5024C-330	Main Housing	2	Cast Iron	SP-81348
FM-5024C-514	Rotor Assembly	4	Plastic	SP-63245
FM-5024C-549	Stator Assembly	4	Cast Iron	SP-56793
FM-5024C-987	Bearing Set	3	Stainless Steel	SP-88474
FM-5024C-791	Shaft	2	Cast Iron	SP-94133
FM-5024C-893	Seal Kit	4	Aluminum	SP-53489
FM-5024C-828	Gasket Set	3	Bronze	SP-31787
FM-5024C-501	Mounting Bracket	1	Bronze	SP-67539
FM-5024C-956	Cooling Fan	2	Cast Iron	SP-29865
FM-5024C-808	Terminal Box	1	Stainless Steel	SP-68514
FM-5024C-993	Nameplate	1	Aluminum	SP-85261
FM-5024C-864	Fastener Kit	3	Aluminum	SP-56802
FM-5024C-934	Thermal Protection	4	Aluminum	SP-16898
FM-5024C-116	Vibration Sensor	4	Steel	SP-74111
FM-5024C-529	Oil Filter	2	Bronze	SP-67415
FM-5024C-464	Cooling Coil	1	Stainless Steel	SP-75811
FM-5024C-702	Impeller	4	Bronze	SP-88826
FM-5024C-571	Volute Casing	4	Aluminum	SP-82991
FM-5024C-389	Mechanical Seal	1	Steel	SP-85595
FM-5024C-706	Shaft Sleeve	1	Aluminum	SP-89297
FM-5024C-127	Coupling Half	3	Stainless Steel	SP-32113

Recommended Spare Parts

Part Number	Description	Recommended Stock	Lead Time
FM-5024C-833	Bearing Set	3-5	4-6 weeks
FM-5024C-813	Seal Kit	1-2	4-6 weeks
FM-5024C-360	Gasket Set	1-2	2-4 weeks
FM-5024C-665	Filter Element	1-2	2-4 weeks
FM-5024C-898	Thermal Fuse	2-3	1-2 weeks
FM-5024C-852	Vibration Sensor	2-3	2-4 weeks
FM-5024C-123	Cooling Fan	2-3	4-6 weeks

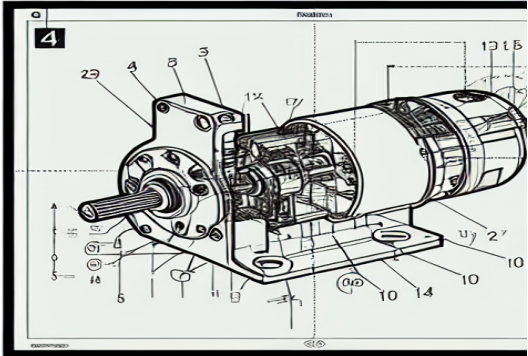
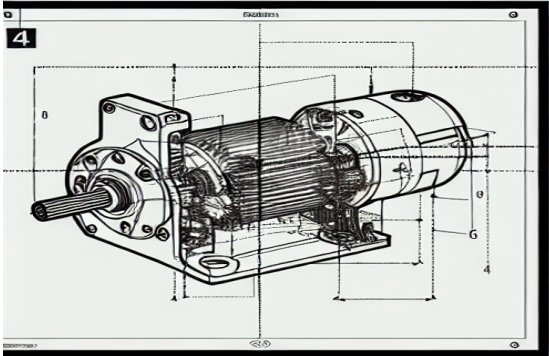
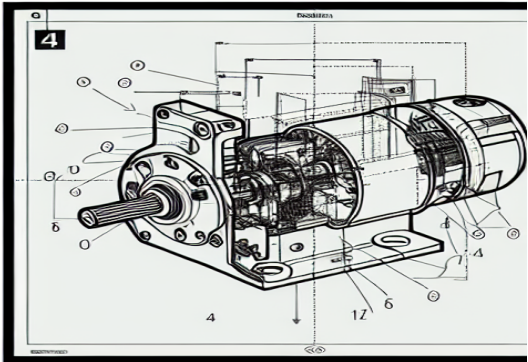
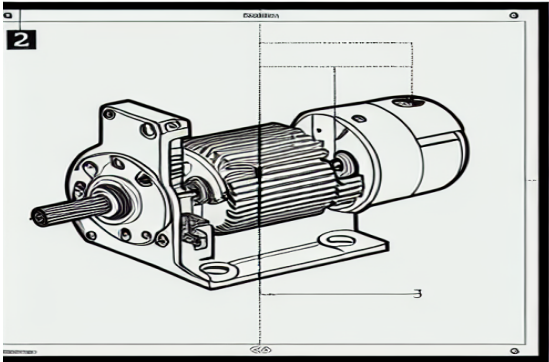
Lubrication Requirements

Component	Lubricant Type	Quantity	Interval
Main Bearings	Oil ISO VG 68	148	Every 1593 hours
Gearbox	Oil ISO VG 220	2	Every 3692 hours
Seals	Silicone Grease	28	Every 568 hours

Refer to the wiring diagrams provided in the appendix. All connections must be made according to local electrical codes and regulations.

Terminal Connections

Terminal	Function	Wire Size	Voltage
L1, L2, L3	Power Input	6 mm ²	690V
PE	Protective Earth	4 mm ²	0V
U, V, W	Motor Output	4 mm ²	Variable
24V	Control Supply	0.75 mm ²	24V DC
GND	Control Ground	0.75 mm ²	0V



The Industrial Motor is compliant with applicable international standards for industrial machinery and electrical equipment. In case of deviation from these standards, these are listed in the declaration of incorporation. The declaration of incorporation is part of the delivery.

Machine Standards

Standard	Description
IEC 60034-1	Rotating electrical machines - Part 1: Rating and performance
IEC 60034-2-1	Rotating electrical machines - Part 2-1: Standard methods for determining losses and efficiency
IEC 60034-5	Rotating electrical machines - Part 5: Degrees of protection provided by the integral design of rotating electrical machines (IP code)
IEC 60034-6	Rotating electrical machines - Part 6: Methods of cooling (IC code)
IEC 60034-30-1	Rotating electrical machines - Part 30-1: Efficiency classes of line operated AC motors (IE code)

Other Standards Used in Design

Standard	Description
IEC 60204-1	Safety of machinery - Electrical equipment of machines - Part 1: General requirements
IEC 61000-6-2	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
IEC 61000-6-4	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
ISO 13849-1:2015	Safety of machinery - Safety related parts of control systems - Part 1: General principles for design
IEC 61508	Functional safety of electrical/electronic/programmable electronic safety-related systems

A. Technical Data Sheets

Additional technical data sheets and detailed specifications are available upon request.

Revision History

Revision History

Revision	Description
S	- Published in release R18.1. The following updates are made in this revision: - Added sections in general procedures. - Safety restructured. - Updated spare part number brake release board (was DSQC513, is DSQC1013). - Information about business portal added.
P	- Published in release R22A. The following updates are made in this revision: - Clarified and added information in mounting instructions for rotating sealings. - Added information about optimization and calibration procedures in calibration chapter. - Replaced article number and name of grease, previously 3376178-2.
X	- Published in release R21B. The following updates are made in this revision: - New tool for replacement of parallel bar. - Updated maintenance procedures. - Added calibration requirements.
V	- Published in release R24A. The following updates are made in this revision: - Added sections in general procedures. - Safety restructured. - Updated spare part number brake release board (was DSQC513, is DSQC1013). - Information about business portal added.
Q	- Published in release R17.2. The following updates are made in this revision: - Updated information about Loctite treated screws. - Removed information about position switches as they are no longer available. - Updated torque specifications for various components.
Y	- Published in release R19B. The following updates are made in this revision: - Clarified and added information in mounting instructions for rotating sealings. - Added information about optimization and calibration procedures in calibration chapter. - Replaced article number and name of grease, previously 3376178-2.
T	- Published in release R20B. The following updates are made in this revision: - Clarified and added information in mounting instructions for rotating sealings. - Added information about optimization and calibration procedures in calibration chapter. - Replaced article number and name of grease, previously 3376178-2.
N	- Published in release R24D. The following updates are made in this revision: - New tool for replacement of parallel bar. - Updated maintenance procedures. - Added calibration requirements.

B. Contact Information

Contact Information

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